

Sulfuric Acid Production Design Project

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Abstract

Due to its various utilities, sulfuric acid is one of the most produced chemical compounds in the world. This has resulted in countless efforts to create the most efficient plants for maximum manufacturing. This report focuses on the single contact process. Although slightly inferior to the double contact process, the single contact process nonetheless, has a conversion rate of 97% of sulfur dioxide in the catalytic convertor, a 2 to 2.5% difference. A small plant can nonetheless produce large amounts of sulfuric acid a day, the plant specifically producing 1166.18 tons/day considering a 3-week downtime for maintenance. This significant quantity results in 128.66 MW of heat released for the overall plant. Such large amounts produced results in more consequences, therefore, environmental, economic, safety, and improvements were explored in detail. The cost to run such a plant is estimated to range from 151.05 million CAD to 257.25 million CAD assuming the plant is based in Quebec.

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1 Introduction

Sulfuric acid is the world's most mass-produced industrial chemical. It consists of a colourless, oily liquid, which is known for its extremely reactive behaviour thanks to it being a strong acid. Although it is a corrosive substance, sulfuric acid is prominent in various industries. Notably, it's a core element in the production of fertilizers, petroleum refining, metal processing and explosives. This makes it a very sought-for substance thanks to its indispensable nature in numerous chemical reactions. Hence, the constant push for innovative techniques in its production is a necessity. This is why the most common modern method of production is the contact process, thanks to its characteristically concentrated sulfuric acid [1].

1.1 History

The production of sulfuric acid was not always as significant as it is in modern times. In fact, it had a modest upbringing during the early to mid-18th century, where it was sold under the form of sodium sulphate decahydrate as a cleaning agent [2]. This limited its demand to a small amount, which meant that its production was minimal. In 1736, Joshua Ward started the small-scale production of this acid with the burning of a mixture of sulfur and potassium nitrate over a shallow layer of water in glass vessels. Then, in 1747, John Roebuck increased the production of the liquid by replacing the glass vessels with large lead chambers on wooden frames [3]. This brought production from only a couple pounds per glass vessel to a hundred pounds per lead chamber [4]. The liquid produced had a concentration of 35-45% sulfuric acid, which could be concentrated through boiling. Demand further rose in 1760 [2], with the newfound application for sulfuric acid as a bleaching agent in the dyeing industry. However, this application was short lived because it got replaced by chlorine. This led to a soar in demand for sulfuric acid because chlorine was mainly made with a combination of it and salt [3]. Therefore, this rise in demand required an increase in the manufacturing capacity, which brought by necessity the creation of the first continuous chemical process.

The lead chamber process can be summarized by the creation of sulfur dioxide by burning sulfur, which is then used to create sulfuric acid. The process consists of major improvements that were made to Roebuck's outline [2]. First, the introduction of Gay-Lussac's towers in 1827 reduced the consumption of potassium nitrate, which acted as a catalyst in the process, by enabling the recovery of nitrogen oxide gases. Furthermore, it increased the sulfuric acid's production by blowing supplemental air into the lead chambers. This was further improved in 1859 by John Glover, whose design enhanced the recovery of nitrogen oxide and heightened the concentration

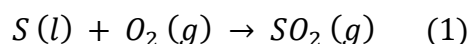
of the final product. Second, producers turned away from the monopolized Sicilian sulfur, which was the primary source of raw material, and transitioned to roasting pyrites. This led to a decrease in the cost of raw materials and returned a byproduct of copper and iron following the sulfur's extraction from the ore. Therefore, these two elements permitted to reduce the cost and increase the capacity of production. In fact, between 1860 and 1900 the production of sulfuric acid tripled in Britain (first industrial country) [4]. This demonstrates the revolution brought by the lead chamber process, which also yielded a higher concentration of 70-78% acid. However, this process was later displaced by the contact process, with its last plant shutting down in 1973 [3].

The contact process is the most common modern sulfuric acid production method [2]. This process was patented in 1831 by Peregrine Philips [5], but it only overtook the previous lead chamber process after World War 1. This is no coincidence as research in chemistry soared during the war, and an important innovation permitted the large-scale operation of this process. In fact, under normal conditions, sulfur dioxide does not bind with oxygen present in the air, which complicates the creation of sulfur trioxide (needed to make sulfuric acid). Therefore, in 1880, Rudolf Knietsch demonstrated that in the presence of a catalyst, sulfur dioxide does bind with the oxygen, forming sulfur trioxide. Two catalysts were found to be suitable: platinum and vanadium oxide [2]. Although platinum is considered as a better catalyst, its high cost and it being prone to poisoning (deactivation) made vanadium oxide the catalyst of choice. However, this only increases the rate of reaction and has no effect on the amount of sulfur trioxide produced [5]. Hence, this enabled the exploitation of the contact process, which yields greater sulfuric acid concentration, reaching up to 98% [2]. The contact process can be divided into two types of plants. The simpler sulfur-burning ones (raw sulfur) and the ones that use pyrites (low-grade sulfur). In this scenario, the contact plant with raw sulfur is explored. It can be generalized into 3 great steps. First, the

generation of sulfur dioxide from raw sulfur that burned in the presence of oxygen [6]. Second, the oxidation of sulfur dioxide into sulfur trioxide with the use of the vanadium pentoxide catalyst. Third, sulfur trioxide is converted to sulfuric acid, which gives the desired product [5].

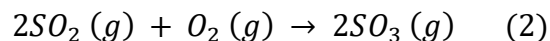
1.2 Background Information

The contact process is a widely used industrial method to produce sulfuric acid. Since the introduction of a catalyst to facilitate the oxidation of sulfur dioxide, the contact process has been the primary method of manufacturing. The process is also cost effective and efficient, making it ideal for large scale production. The first step is the extraction of elemental sulfur. There are various ways to accomplish this, but the most common is the Frasch process [7], which involves superheating water and injecting it into underground sulfur deposits, thus forcing the sulfur upwards where it can be collected. Once the elemental sulfur has been extracted, the contact process begins with the oxidation of the elemental sulfur to produce sulfur dioxide.



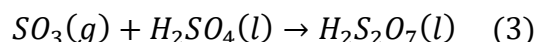
The reaction will occur at 300 °C, and thus the sulfur is heated prior to the reactor in a sulfur melter, converting the solid sulfur into a liquid. Due to the exothermic nature of the reaction, cooling jackets are used on the sides of the reactor to remove heat and ensure that the reaction proceeds. Due to the excess of air used to fuel this reaction, it is assumed that 100% of the sulfur is converted in this reaction.

The second step involves the oxidation of sulfur dioxide into sulfur trioxide. The reaction is highly exothermic and requires a catalyst, vanadium oxide, to lower the activation energy and drive the reaction forward.

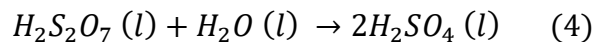


The temperature and pressure are carefully controlled by heat exchangers and compressors to ensure that they remain at the optimal temperatures and pressures according to LeChatelier's principle [8]. The conversion of sulfur dioxide in this reaction in industry is roughly 97%, which can be attributed to the fact that this is an equilibrium-limited reaction, and the sulfur trioxide is not removed from the reactor until the full reaction is completed.

The third step of the process is the absorption of sulfur trioxide into an industry standard 98 weight percent concentrated, liquid sulfuric acid which produces oleum [9].



Oleum is a necessary intermediate, as direct absorption of sulfur trioxide and water would result in an unstable reaction and produce an acid mist. Due to the nature of a gas-liquid reaction, some of the gas is not able to be absorbed into the liquid and therefore exits the reactor as off gas. The conversion of the sulfur trioxide is roughly 97% [10]. The next step in the contact process involves the conversion of this intermediate, oleum, into concentrated sulfuric acid with the addition of liquid water.



It is important to note that the reaction produces two times as much sulfuric acid as inlet oleum. This process is highly efficient and has a very high conversion of oleum in both an idealistic and realistic context. The sulfuric acid will then be diluted to industry standard (98 wt% sulfuric acid) in any excess water, and either recycled or stored.

2 Process Overview

2.1 Assumptions

To ensure a valid process description, we must define some assumptions that will serve as guidelines throughout. The best way to gather information on such processes is using real-life plants that are in use near us. It is considered that a small sulfuric acid plant will only operate under a single contact process. This process assumes that a minimum quantity of 1166.18 tons/day will be produced [11]. In a year's time, assuming 3 weeks of the year the production process is turned off due to maintenance, 400,000 tons of sulfuric acid will be produced. It is important to note that we assume this is 400,000 tons of pure sulfuric acid, not including the 2-weight percent of water that it must be diluted in.

We will assume ideal conditions for all gases and liquids, as well as steady flow rates, and the heat exchangers pressure changes are insignificant enough for which they can be ignored. The process begins with the isobaric melting of sulfur. The sulfur is isothermally pumped to a furnace where an isothermally pumped source of dry air reacts. It is assumed that the air used is pre-dried. This process is usually achieved by utilizing a strong acid to remove the moisture present in the air. Sulfuric acid is used by many industries as its strong hygroscopic properties act as an efficient drying agent [12]. The furnace undergoes isobaric combustion. The combustion of sulfur dioxide is the only reaction occurring for which it is ideal, and an excess of air is supplied for the combustion of sulfur dioxide to be 100%. Although there remains a significant quantity of oxygen going into the catalytic convertor from the furnace, another stream of dry air is fed to the device to ensure a complete oxidization of sulfur dioxide into sulfur trioxide [13].

The one catalytic bed convertor is used to produce sulfur trioxide therefore the conversion rate is not as high as one with two to four. The conversion of sulfur dioxide for the single contact catalytic converter is assumed to be 97% [14]. The catalyst, vanadium oxide, is introduced once into the catalytic convertor as it has a service life of approximately 5-10 years [15]. In that regard, it is assumed that a constant supply of the catalyst is not necessary. The sulfur trioxide undergoes isobaric cooling and is then passed through an absorption tower to react with 98 wt% concentrated sulfuric acid (in which the 2% is water). This small presence of water in the sulfuric acid is assumed to not react with the sulfur trioxide. It is assumed that 97% of the sulfuric trioxide introduced transforms into oleum, and 100% of the sulfuric acid is consumed in the reaction. The remaining 3% exits the tower as tail gas [16]. This reaction occurs at a temperature ranging between 50 to 70°C as a much higher temperature would favour the production of an acid mist which is the reaction between sulfur trioxide and water. This would mean that the sulfur trioxide would not be soluble in the sulfuric acid, ultimately jeopardizing the wanted outcome. Although realistically, even at a system temperature in that range, a small quantity of acid mist would be produced, however, since there is such little water present, we neglect this reaction and assume it does not occur. The oleum is cooled with a constant pressure and is reacted with water inside a dilution tank. This process creates concentrated sulfuric acid. It has a 100% conversion rate both in an ideal and realistic context and has a yield of 30% [17].

2.2 Process Flow Diagram

Table 1. Description of Devices Presented in the Process Flow Diagram

Devices	Utility	Devices	Utility
T-101	Sulfur Melter	T-302	Water Tank
P-101	Molten Sulfur Pump	P-301	Water Pump
C-101	Air Compressor	E-301	Oleum Cooler
R-101	Sulfur Furnace	R-302	Oleum Dilution Tower
E-201	Sulfur dioxide Cooler	E-302	Sulfuric Acid Cooler
R-201	Catalytic Convertor	E-303	Sulfuric Acid Heater
E-202	Sulfur trioxide Cooler	T-301	Sulfuric Acid Tank
R-301	Oleum Absorption Tower	-	-

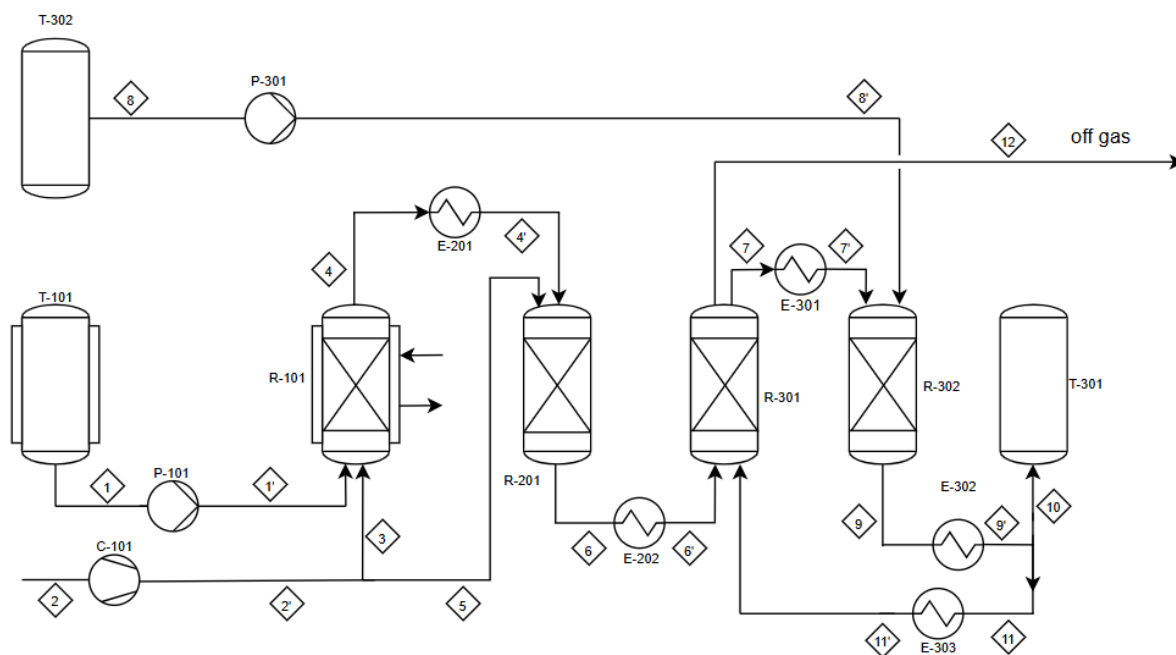


Figure 1. Process Flow Diagram for the Proposed Production of Sulfuric Acid Through the Single Contact Process Method

Process Flow Diagram

The molten sulfur passes through a pump (P-101) and combines with dry air inside the furnace (R-101). The combustion of elemental sulfur with an excess of oxygen from the air produces sulfur dioxide. The gas leaves the furnace with a temperature of 1070°C [18]. The gas is further cooled through a heat exchanger (E-201) to 425°C [19]. A stream of air reacts with sulfur dioxide and vanadium oxide inside the catalytic converter (R-201) to form sulfur trioxide. The catalytic converter consists of 1 reacting bed which reacts sulfur dioxide and excess air, catalyzed with vanadium oxide to form sulfur trioxide. The sulfur trioxide leaves at a temperature of 403°C, cooled down to 70°C, and fed into an absorption tower (R-302) where it reacts with a 65°C sulfuric acid stream to produce oleum. Tail gas which mainly consists of nitrogen exits the process. The oleum leaves at a temperature of 67°C, is cooled to 50°C, and fed into a dilution tower (R-302) where water reacts with the oleum to produce sulfuric acid [20]. The sulfuric acid leaves the tower with a temperature of 40°C and is cooled to 25°C before being stored (T-301). Pressures for each stage was not mentioned as it never tends to deviate from a range of 1 to 2 bars. Although an increase in pressure could benefit the equilibria, favouring the production of product, the cost of maintaining an elevated pressure beyond that of atmospheric is pointless. Many plants favour optimizing temperature ranges, ideally striking a balance between 400 and 450°C in the catalytic converter as the catalyst used, vanadium oxide, is efficient enough [17].

2.3 Stream Table

Table 2. Properties for Streams 1 Through 12

Streams	Temperature (K)	Pressure (bar)	$\dot{m}_i$ (kg s ⁻¹)	$\dot{n}_i$ (mol s ⁻¹)
1	433.15	1.2	4.688	146.288
1'	433.15	1.8	4.688	146.288
2	298.15	1	59.673	2067.596
2'	298.15	1.4	59.673	2067.596
3	298.15	1.4	40.222	1393.526
4	1343.15	1.5	44.836	1392.316
4'	698.15	1.5	44.836	1392.316
5	298.15	1.4	19.47	675.28
6	676.15	1.3	64.353	1996.648
6'	343.15	1.3	64.353	1996.648
7	340.15	1.2	24.795	152.89
7'	323.15	1.2	24.795	152.89
8	298.15	1	2.755	152.89
8'	298.15	1.55	2.755	152.89
9	313.15	1	27.55	305.78
9'	298.15	1	27.55	305.78
10	298.15	1	13.775	152.89
11	298.15	1	13.775	152.89
11'	338.15	1.25	13.775	152.89
12	363.15	1.1	53.344	1859.008

2.4 Mass Balances

We can begin with a degree of freedom analysis around each reactor.

Degrees of freedom:

$$DoF = \#parameters - \#equations \quad (1)$$

R-101

6 parameters ($\dot{n}_{2,O_2}, \dot{n}_{1,S}, \dot{n}_{2,N_2}, \dot{n}_{4,O_2}, \dot{n}_{4,O_2}, \dot{n}_{4,SO_2}$)

6 equations ($N_2: O_2$ ratio, heat balance, mol balance, conversion of sulfur, conversion of oxygen,

1 reaction)

$$6 \text{ parameters} - 6 \text{ equations} = 0 \text{ degrees of freedom}$$

R-201

9 parameters ($\dot{n}_{4,O_2}, \dot{n}_{4,N_2}, \dot{n}_{5,O_2}, \dot{n}_{5,N_2}, \dot{n}_{4,SO_2}, \dot{n}_{6,O_2}, \dot{n}_{6,N_2}, \dot{n}_{6,SO_2}, \dot{n}_{6,SO_3}$)

6 equations

(heat balance, 2 mol balances, conversion of SO_2 , conversion of O_2 , 1 reaction)

$$9 \text{ parameters} - 6 \text{ equations} = 3 \text{ degrees of freedom}$$

R-301

13 parameters ($\dot{n}_{6,O_2,N_2,SO_2,SO_3}, \dot{n}_{11,H_2O,H_2SO_4}, \dot{n}_{12,N_2,O_2,SO_2,SO_3}, \dot{n}_{7,H_2S_2O_7,H_2O}$)

6 equations (3 mol balances, heat balance, 1 reaction, conversion of SO_3)

$$13 \text{ parameters} - 6 \text{ equations} = 7 \text{ degrees of freedom}$$

R-302

5 parameters ($\dot{n}_{8,H_2O}$, $\dot{n}_{7,H_2O,H_2S_2O_7}$, $\dot{n}_{9,H_2O,H_2SO_4}$)

4 equations (*mol balance, heat balance, 1 reaction, conversion of $H_2S_2O_7$*)

$$5 \text{ parameters} - 4 \text{ equations} = 1 \text{ degree of freedom}$$

Accounting for the ties, (parameters that are used twice), we have a total of:

$$11 \text{ degrees of freedom} - 10 \text{ ties} = 1 \text{ degree of freedom}$$

Thus, we will have to create an assumption to solve the system. We will assume that the sulfuric acid splits evenly between streams 10 (final product stream) and 11 (recycle stream). Now we may proceed with the material balance.

Material balance

To begin with the material balance, we convert the goal of 400,000 metric tons of sulfuric acid per 49 weeks to kilograms per second. This conversion simplifies calculations further down the line.

$$\left[\frac{400,000 \text{ tonnes}}{49 \text{ weeks}} \right] \left[\frac{1000 \text{ kg}}{1 \text{ tonne}} \right] \left[\frac{1 \text{ week}}{604800 \text{ seconds}} \right] = 13.497 \sim 13.5 \frac{\text{kg } H_2SO_4}{s}$$

Then, using the molar mass of sulfuric acid to be $98.08 \frac{g}{mol}$, we can calculate the molar flow rate.

$$\left[13.5 \frac{\text{kg } H_2SO_4}{s} \right] \left[\frac{1000 \text{ g}}{\text{kg}} \right] \left[\frac{1 \text{ mol } H_2SO_4}{98.08 \text{ g}} \right] = \frac{137.64 \text{ mol } H_2SO_4}{s}$$

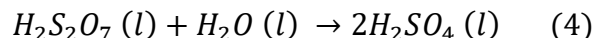
This molar flow rate is the molar flow rate of H_2SO_4 in stream 10, or our final product stream.

Sulfur Balance

Now, working backwards, we can see that this stream is not actually the full amount of sulfuric acid that reactor R-302 produces. An unknown amount is recycled into reactor R-301, where it reacts with sulfur trioxide to produce oleum. The assumption is made that the recycle stream and final product stream split equimolarly. Thus, it can be assumed that the outlet stream of sulfuric acid reactor from R-302 (stream 9) is equal to two times the final product stream.

$$\dot{n}_{9,H_2SO_4} = 2\dot{n}_{10,H_2SO_4} = 275.28 \frac{\text{mol } H_2SO_4}{s}$$

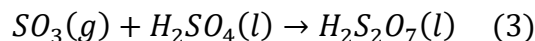
Now, using the assumed conversion of oleum in reactor R-302, which is 100%, we can calculate the number of mols of oleum at the inlet by analyzing reaction 4.



With a conversion of 100% and no side reactions, it is known that the number of moles of sulfuric acid coming out is equal to twice the number of moles of oleum going in, thus:

$$\dot{n}_{7,H_2S_2O_7} = \frac{1}{2} \dot{n}_{9,H_2SO_4} = 137.64 \frac{\text{mol } H_2S_2O_7}{s}$$

Moving further backward to R-301, we note the conversion of sulfur trioxide in reaction 3 is equal to 97%. In other words, the difference between the outlet and inlet streams divided by the inlet streams is equal to 0.97. Thus, we can infer that not all the sulfur trioxide is consumed in this reaction, and some must exit the tower as off gas.



If the sulfur trioxide has a conversion of 97 it can be calculated that the inlet stream of sulfur trioxide is:

$$\dot{n}_{6,SO_3} = \frac{\dot{n}_{7,H_2S_2O_7}}{0.97} = 141.9 \frac{\text{mol } SO_3}{s}$$

Our assumptions state that although the conversion of sulfur trioxide is 97%, the conversion of sulfuric acid in this reactor is 100%. This assumption is made on the fact that there is too little water in the diluted sulfuric acid stream (2 wt% water), such that the reaction between sulfur trioxide and liquid water does not occur. Since the sulfur trioxide and sulfuric acid react in a one-to-one ratio, we can infer that the sulfuric acid entering the reactor is:

$$\dot{n}_{11,H_2SO_4} = 0.97\dot{n}_{SO_3} = 137.64 \frac{\text{mol } H_2SO_4}{s}$$

This aligns with our previous assumption that stream 9 splits equimolarly. Now, looking at R-201 we know our desired sulfur trioxide molar flow rate to be 141.9 mol/s. Our assumptions state that the conversion sulfur dioxide in reaction 2 is also 97%, and we can use this information to find both the outlet and inlet flow rates of sulfur dioxide. Using the same logic as in R-301,

$$\dot{n}_{4,SO_2} = \frac{\dot{n}_{6,SO_3}}{0.97} = 146.288 \frac{\text{mol } SO_2}{s}$$

The sulfur dioxide which does not react with the oxygen remains in the stream as a gas and will exit the system later in stream 12. With the molar flowrate of sulfur dioxide in stream 4, we know that it must be equal to the total mol of liquid sulfur going into R-101, as with an excess of oxygen, we can assume that the sulfur gets completely converted into sulfur dioxide. Thus:

$$\dot{n}_{1,S} = \dot{n}_{4,SO_2} = 146.288 \frac{\text{mol } S}{s}$$

Oxygen Balance

To ensure that both of our oxidation reactions react as completely as possible, an excess of oxygen is needed in each of these reactors. Thus, we have decided to double the moles of oxygen required in each reactor. In reactor 1, we note 146.288 moles of sulfur at the inlet. Thus, the same amount of moles of O_2 will react with the sulfur; thus, the inlet stream of oxygen will be:

$$\dot{n}_{3,O_2} = 2\dot{n}_{1,S} = 292.576 \frac{\text{mol } O_2}{s}$$

The same logic can be applied to the second reactor; however, we note that the stoichiometric quantity of oxygen is exactly half of the amount of product. Thus:

$$\dot{n}_{5,O_2} = (2) \left(\frac{1}{2} \right) \dot{n}_{6,SO_3} = 141.9 \frac{\text{mol } O_2}{s}$$

We can combine these two streams to acquire the total moles of oxygen at stream 2 to be:

$$\dot{n}_{2,O_2} = \dot{n}_{3,O_2} + \dot{n}_{5,O_2} = 434.476 \frac{\text{mol } O_2}{s}$$

Since oxygen by itself does not react in any other reactor, any excess oxygen leaves the process as off gas in stream 12. The oxygen from water is discussed in the section below.

Water Balance

Since sulfuric acid is difficult to contain as a pure species, it is often diluted in water to improve handling ability. Typically, sulfuric acid comprises 98 wt%, while water makes up 2 wt%. In our process, however we want slightly stronger sulfuric acid and have chosen a 99.5 wt% acid and 0.5 wt% water. From this we can calculate the total moles of water that must exit R-302 to maintain

this concentration. First, we must calculate the desired mass of both species, then convert the mass of water to moles.

$$\dot{m}_{9,H_2SO_4} = \left[275.28 \frac{\text{mol } H_2SO_4}{s} \right] \left[\frac{98.08 \text{ g}}{\text{mol}} \right] \left[\frac{1 \text{ kg}}{1000 \text{ g}} \right] = 26.999 \sim 27 \frac{\text{kg } H_2SO_4}{s}$$

The desired wt% of water is 2%, thus:

$$\dot{m}_{9,H_2O} = \frac{\dot{m}_{9,H_2SO_4}}{0.98} - \dot{m}_{9,H_2SO_4} = 0.55 \frac{\text{kg } H_2O}{s}$$

Converting to moles:

$$\dot{n}_{9,H_2O} = \left[0.55 \frac{\text{kg } H_2O}{s} \right] \left[\frac{1000 \text{ g}}{1 \text{ kg}} \right] \left[\frac{1 \text{ mol}}{18.02 \text{ g}} \right] = 30.5 \frac{\text{mol } H_2O}{s}$$

The water concentration will split equally between streams 10 and 11, so:

$$\dot{n}_{10,H_2O} = \dot{n}_{11,H_2O} = \frac{1}{2} \dot{n}_{9,H_2O} = 15.25 \frac{\text{mol } H_2O}{s}$$

Our previous assumption stated that the water in stream 11 does not react with the sulfur trioxide in R-301, thus, the amount of water in stream 7 is equal to:

$$\dot{n}_{7,H_2O} = \dot{n}_{11,H_2O} = 15.25 \frac{\text{mol } H_2O}{s}$$

Now we determine the amount of water we need to add to reactor 4 we must consider several things. First, the stoichiometric quantity of water reacts one-to-one with the amount of oleum. With an oleum conversion rate of 100%, we know the amount of moles of water that will react with the oleum is equal to the amount of oleum in the reactor, 137.64 moles per second. Since we want an excess of 30.5 moles of water for our desired concentration of sulfuric acid, this means that the total water quantity entering the reactor must be equal to following expression:

$$\dot{n}_{in,H_2O} = 137.64 \frac{\text{mol}}{\text{s}} + 30.5 \frac{\text{mol}}{\text{s}} = 168.14 \frac{\text{mol } H_2O}{\text{s}}$$

We already have 15.25 moles entering from stream 11, thus we need:

$$168.14 \frac{\text{mol } H_2O}{\text{s}} - 15.25 \frac{\text{mol } H_2O}{\text{s}} = 152.89 \frac{\text{mol } H_2O}{\text{s}}$$

Entering from stream 8.

Off Gas

Stream 12, the off-gas stream is comprised of oxygen, nitrogen, sulfur dioxide and sulfur trioxide.

Nitrogen is not used in any reaction, and thus the amount in this stream is equal to the amount coming into the system, which can be calculated by the 21:79 molar ratio of oxygen to nitrogen in air. The same ratio applied to streams 3 and 5.

$$\dot{n}_{2,N_2} = \dot{n}_{2,O_2} \left[\frac{79}{21} \right] = 1633.12 \frac{\text{mol } N_2}{\text{s}}$$

$$\dot{n}_{12,N_2} = \dot{n}_{2,N_2} = 1633.12 \frac{\text{mol } N_2}{\text{s}}$$

Oxygen is in excess in reactions 1 and 2, and is not used in further reactions and thus the excess leaves in this stream as well. It is calculated by subtracting the amounts used in reactions 1 and 2 from the initial amount.

$$\dot{n}_{12,O_2} = \dot{n}_{2,O_2} - \frac{1}{2} \dot{n}_{5,O_2} - \frac{1}{2} \dot{n}_{3,O_2} = 217.24 \frac{\text{mol } O_2}{\text{s}}$$

Sulfur dioxide has a 97% conversion to sulfur trioxide, and the rest remains in the stream until it leaves in stream 12. The same process can also be done for sulfur trioxide.

$$\dot{n}_{12,SO_2} = 0.03\dot{n}_{4,SO_2} = 4.388 \frac{mol SO_2}{s}$$

$$\dot{n}_{12,SO_3} = 0.03\dot{n}_{7,SO_3} = 4.26 \frac{mol SO_3}{s}$$

The masses of each stream can be found by the equation:

$$\dot{m} \left[\frac{kg}{s} \right] = \dot{n} \left[\frac{mol}{s} \right] \times M \left[\frac{g}{mol} \right] \times \left[\frac{1 kg}{1000 g} \right]$$

Mass balances for each reactor are shown below.

R-101

Table 3. Mass Balance for Reactor 101

Reactor 1			
	In	Out	Difference
total mass (kg/s)	44.91	44.836	0.074

Reactor 1's inlet stream for total mass is calculated by adding the total mass of streams 1 and 3.

The outlet stream is calculated by the total mass of the fourth stream.

R-202

Table 4. Mass Balance for Reactor 202

Reactor 2			
	In	Out	Difference
total mass (kg/s)	64.306	64.353	0.047

Reactor 2's inlet stream is calculated as the total mass of streams 4 and 5, and the outlet stream is the total mass of stream 6.

R-301

Table 5. Mass Balance for Reactor 301

Reactor 3			
	In	Out	Difference
total mass (kg/s)	78.128	78.139	0.011

Reactor 3's inlet stream is calculated as the total mass of streams 6 and 11, and the outlet stream is the total mass of streams 12 and 7.

The composition of oleum consists of sulfur trioxide and sulfuric acid (31% and 69% respectively). These values fall into the most commonly produced weight percentage in terms of industry standards [17].

R-302

Table 6. Mass Balance for Reactor 302

Reactor 4			
	In	Out	Difference
total mass (kg/s)	27.55	27.55	0.00

Reactor 4's inlet stream is calculated as the total mass of streams 7 and 8, and the outlet stream is calculated as the total mass of stream 9.

The mass balances for each reactor prove that all mass is conserved within the process. The small

human differences between the inlet and outlets at reactors 1, 2 and 3 can be attributed to uncertainty and human errors while rounding.

Total mol balance:

We can also conduct a total mol balance to ensure that the amount of moles of each element are conserved throughout the process.

Table 7. Total Mass Balance

Mol Balances	In	Out	Difference
S	146.288	146.288	0
H	305.78	305.78	0
O	1021.842	1021.846	0.004

The differences are all relatively zero, and thus we can be sure that each element is conserved. An excel spreadsheet can be found in the appendix which provides deeper insight onto our calculations and balances.

2.5 Heat Balances

Table 8. Hyperbolic Heat Capacity Constants [Perry's Textbook]

Species	A [J (K·mol) ⁻¹]	B [J (K·mol) ⁻¹]	C [J (K·mol) ⁻¹]	D [J (K·mol) ⁻¹]	E [J (K·mol) ⁻¹]
*S _(l)	-4540.97	26065.6	-55520.7	42012.2	54.58860
O _{2(g)}	29.103	10.04	2526.5	9.356	1153.8
SO _{2(g)}	33.375	25.864	932.8	10.88	423.7
SO _{3(g)}	33.408	49.677	873.22	28.563	393.74
*V ₂ O _{5(s)}	45.97044	14.46601	0.619521	-0.177508	-0.420735
*H ₂ SO _{4(l)}	79.837	-	-	-	-
*H ₂ S ₂ O _{7(l)}	113	-	-	-	-
H ₂ O _(l)	33.363	26.790	2610.5	8.896	1169
N _{2(g)}	29.105	8.6149	1701.6	0.10347	909.79

*In this table, the linear heat capacity constants from NIST are used for sulfur and vanadium pentoxide [21]. However, NIST refers to vanadium pentoxide as vanadium oxide, which represents the same thing according to numerous sources [22]. Furthermore, sulfuric acid and oleum do not have any data on how their heat capacity constants change with temperature. Therefore, they will be assumed to be constant at the values marked in the “A” column of the table.

Table 8 will be used to calculate how the heat capacity constants vary for the species based on the temperature changes. This is mainly achieved through Excel that utilizes the hyperbolic function. It allows for the calculation of the in and out enthalpy.

Table 9. Standard Heats of Formation for given species [NIST Database and PubChem]

Species	$\Delta H^{\circ}_{f,i}$ [kJ mol ⁻¹]	MW(g mol ⁻¹)
S _(l)	1.85	32.065
O _{2(g)}	0	32.00
SO _{2(g)}	-296.84	64.064
SO _{3(g)}	-395.77	80.063
V ₂ O _{5(s)}	-1550.6	181.88
H ₂ SO _{4(l)}	-909.3	98.079
H ₂ S ₂ O _{7(l)}	-1272.4	178.14
H ₂ O _(l)	-285.83	18.02
N _{2(g)}	0	28.02

The heat of formation method was used to calculate energy balances:

$$Q = \Delta H = \Sigma(\dot{n}_{i,out} \cdot h_{i,out}) - \Sigma(\dot{n}_{i,in} \cdot h_{i,in}) \quad (2)$$

The enthalpy of each species is calculated by the following equation:

$$h_i(T) = \Delta H^{\circ}_{f,i} + \int_{T_0}^{T_{final}} C_{p,i} \cdot dT \quad (3)$$

Where each species heat capacities can be found using the following polynomial equations:

For linear species:

$$C_p(T) = A + BT + CT^2 + DT^3 + \frac{E}{T^2} \quad (4)$$

The integrated form of Equation 4 is:

$$\int_{T_{initial}}^{T_{final}} Cp(T) = AT + \frac{B}{2}T^2 + \frac{C}{3}T^3 + \frac{D}{4}T^4 - \frac{E}{T} \quad (5)$$

For hyperbolic species, the heat capacity is expressed by:

$$Cp(T) = A + B\left(\frac{\frac{C}{T}}{\sinh \frac{C}{T}}\right)^2 + D\left(\frac{\frac{E}{T}}{\cosh \frac{E}{T}}\right)^2 \quad (6)$$

Sulfur Furnace (R-101)

Table 10. Calculated data for the sulfur furnace

Species	T _{in}	T _{out}	$\dot{n}_{i, in}$	$h_{i, in}$	$\dot{n}_{i, out}$	$h_{i, out}$
	(K)	(K)	(mol s ⁻¹)	(kJ mol ⁻¹)	(mol s ⁻¹)	(kJ mol ⁻¹)
S _(l)	433.15	-	146.288	1.39245	0	1.85
O _{2(g)}	298.15	1343.15	292.576	0	146.288	34.8944
SO _{2(g)}	-	1343.15	0	-296.84	146.288	-243.335
Q (MJ/s)	-30.696					

From Table 10, it is understood that the reaction occurring in the sulfur furnace produces 30.696 MW of energy. This means that a substantial amount of cooling is required to maintain the reactor at a suitable temperature for the exothermic reaction to keep occurring.

In this case the Cp constant of S is calculated using equation 5:

$$\int_{298}^{433} Cp(T) = -4540.97(433.15 - 298.15) + \frac{26065.6}{2}(433.15 - 298.15)^2$$

$$- \frac{55520.7}{3}(433.15 - 298.15)^3 + \frac{42012.2}{4}(433.15 - 298.15)^4$$

$$- \frac{54.58860}{T(433.15 - 298.15)}$$

$$Cp(433.15) = -457.554 \text{ J mol}^{-1}$$

Catalytic Convertor (R-201)

Table 11. Calculated data for the catalytic convertor

Species	T _{in}	T _{out}	$\dot{n}_{i, \text{ in}}$	$h_{i, \text{ in}}$	$\dot{n}_{i, \text{ out}}$	$h_{i, \text{ out}}$
	(K)	(K)	(mol s ⁻¹)	(kJ mol ⁻¹)	(mol s ⁻¹)	(kJ mol ⁻¹)
SO _{2(g)}	676.15	698.15	146.288	-279.514	4.388	-278.397
O _{2(g)} (S5)	298.15	698.15	141.9	0	70.95	12.4324
O _{2(g)} (S4')	676.15	698.15	146.288	11.7090	146.288	12.4324
SO _{3(g)}	-	698.15	0	-395.77	141.9	-370.906
Q (MJ/s)	-11.973					

From Table 11, it is understood that the reaction occurring in the catalytic converter produces 11.973 MW of energy. This means that a substantial amount of cooling is required to maintain the reactor at a suitable temperature for the exothermic reaction to keep occurring. Furthermore, the catalyst is considered as an observer and does not produce any heat or side reactions.

In this case, SO₂ h_{in} is calculated by first using equation 6 with the coefficients from table 9:

$$Cp(T) = 33.375 + 25.864\left(\frac{\frac{932.8}{T}}{\sinh\frac{932.8}{676.15}}\right)^2 + D\left(\frac{\frac{423.7}{T}}{\cosh\frac{423.7}{676.15}}\right)^2; \quad Cp(676.15) = 17.326 \text{ kJ mol}^{-1}$$

Then, from equation 3 with the heat of formation from table 9:

$$h_{in}(T) = -296.84 + 17.326; \quad h_{in}(676.15) = -279.514 \text{ kJ mol}^{-1}$$

After finding both h_{in} and h_{out} by using equation 2 the heat can be calculated Q :

$$Q = (4.388 \cdot -278.397 + 70.95 \cdot 12.4324 + 146.288 \cdot 12.4324 + 141.9 \cdot -370.906) \\ - (146.288 \cdot -279.514 + 146.288 \cdot 11.7090)$$

$$Q = -11.973 \text{ MJ/s (1 MW = 1 MJ/s)}$$

Oleum Absorption Tower (R-301)

The heat capacity value of 98% sulfuric acid was found to be 79.837 J/(mol*K) at room temperature (25°C) and it is assumed that it stays constant throughout [23]. Although a significant assumption, no further data was found for the temperatures used in the process.

While, the heat capacity for oleum (H₂S₂O₇) was found to be 113 J/(mol*K) at room temperature and is also assumed to remain constant [24]. The lack of data forces this assumption to be made.

Table 12. Calculated data for the oleum absorption tower

Species	T _{in}	T _{out}	$\dot{n}_{i, in}$	$h_{i, in}$	$\dot{n}_{i, out}$	$h_{i, out}$
	(K)	(K)	(mol s ⁻¹)	(kJ mol ⁻¹)	(mol s ⁻¹)	(kJ mol ⁻¹)
SO _{3(g)}	343.15	-	141.9	-393.415	0	-395.77
H ₂ SO _{4(l)}	338.15	-	137.64	-906.107	0	-909.3
H ₂ S ₂ O _{7(l)}	-	340.15	0	-1272,4	137.64	-1267.65
Q (MJ/s)	-50.0976					

From Table 12, it is understood that the reaction occurring in the oleum absorption tower produces 50.0976 MW of energy. This means that a substantial amount of cooling is required to maintain the reactor at a suitable temperature for the exothermic reaction to keep occurring.

Oleum Dilution Tower (R-302)

As previously mentioned, the C_p values of sulfuric acid and oleum were found by making a significant assumption. For the dilution tower, the heat capacity values are assumed to remain constant. Considering the relatively low temperature changes, the C_p values would realistically not change much.

Table 13. Calculated data for the oleum dilution tower

Species	T_{in} (K)	T_{out} (K)	$\dot{n}_{i, in}$ (mol s ⁻¹)	$h_{i, in}$ (kJ mol ⁻¹)	$\dot{n}_{i, out}$ (mol s ⁻¹)	$h_{i, out}$ (kJ mol ⁻¹)
H₂O_(g) (S7')	323.15	313.15	2.81	-284.989	2.81	-285.326
H₂O_(g) (S8')	298.15	313.15	140.448	-285.83	2.81	-285.326
H₂S₂O_{7(l)}	323.15	-	137.64	-1269.58	0	-1272.4
H₂SO_{4(l)}	-	313.15	0	-909.3	275.28	-908.102
Q (MJ/s)	-35.8966					

From Table 13, it is understood that the reaction occurring in the oleum distribution tower produces 35.8966 MW of energy. This means that a substantial amount of cooling is required to maintain the reactor at a suitable temperature for the exothermic reaction to keep occurring.

Overall

The sulfuric acid plant would release 128.66 MW of energy if it is assumed that only the reactor units are considered. This is a fair assumption as the course material focuses on reactors. Furthermore, this means that 128.66 MW of cooling is required to maintain the processes at the required temperatures.

3 Important Considerations

3.1 Vanadium Oxide

The inclusion of a catalyst inside the catalytic converter, R-201, is essential for the operation of our plant. This substance ensures the forward progression of sulfur dioxide into sulfur trioxide. The specific catalyst used, vanadium oxide, is not designed to be replaced frequently and is meant to be replaced every 5-10 years [25]. However, making sure it is in proper condition to continue aiding the reaction is crucial. Therefore, the catalyst will be inspected to ensure that it functions properly during the 3-week downtime the plant takes once every year. To maximize the efficiency and longevity of the catalyst, we must ensure that the temperature inside the catalytic converter never changes from a range of 400 to 450°C. Too much heat might destroy its structure rendering it useless. Too little heat makes the reaction not progress fast enough, creating sulfur trioxide condensation, an unwanted byproduct. Ensuring the catalyst has enough surface area inside the catalytic converter facilitates the reaction [26]. Vanadium oxide will be purchased from BASF, a catalyst company based in Germany. Specifically, their Quattro O4-110 catalyst, which comes in the form of star rings. They can be stored in a covered and dry area and will be stored inside the facility [27].

3.2 Safety

Several safety concerns must be considered in the production of sulfuric acid through the contact process to protect plant personnel and equipment. Sulfuric acid is a strong, corrosive acid that can cause severe chemical burns when in contact with the skin [28]. Sulfuric acid is handled in both the absorption (R-301) and dilution (R-302) stages. Operators working in these areas must wear Class B personal protective equipment (PPE), including gloves, face shields, and a one or two-piece acid suit to prevent sulfuric acid contact with skin or eyes [17]. Additionally, safety showers and eyewash stations must be installed near all acid-handling zones in case of exposure [9]. Sulfuric acid is very corrosive thus storage safety is a key consideration. The sulfuric acid storage tank (T-301) should follow NACE International Standard SP 0294-2006, by using stainless or mild steel to slow corrosion [17]. However, sulfuric acid overtime will react with steel, generating hydrogen gas, which is a highly flammable substance that may be explosive when in contact with air. Therefore, sulfuric acid must be stored in an open ventilation system constructed from acid-resistant materials such as CPVC. Hydrogen detection sensors can provide additional safety as well as additional regular inspections [9]. Storage units have the possibility of generating hydrogen gas, therefore, storage tanks should be in accessible areas for emergency services. Additionally, fire suppression systems must be in place to address the risk of flammable or explosive materials [29].

Another consideration that must be considered in the plant is the formation of sulfuric acid mist, especially during the absorption of SO_3 in R-301. Mist can form when sulfur trioxide reacts with water vapor. Sulfuric acid mist is classified as a Category 1 carcinogen (IARC), and inhalation can cause severe respiratory irritation and long-term health risks [9]. To manage this, absorption towers should be equipped with mist eliminators, and proper ventilation in areas with potential for

mist release should be emplaced, such as near R-301 and R-302, respiratory protection is required for all personnel [30].

Vanadium oxide is used as a catalyst to convert SO_2 to SO_3 . While the catalyst is long-lasting and can be recycled for up to 10 years, it is highly toxic if swallowed and can be fatal if inhaled. Thus, maintenance involving the catalyst must, therefore, include sealed handling systems, local exhaust ventilation, and respiratory PPE. Oleum handled in the absorption tower R-201 can cause severe burns when exposed to the skin, may be fatal if inhaled, and is a carcinogen [17]. The area which holds the absorption tower R-201 must be an open ventilation system and it is suggested that a leak detection system should be in place to prevent leakage and exposure to the highly toxic chemical [28].

3.3 Economics

When considering the cost and revenue of production, it is important to note there are frequent changes in market values of both raw materials and the cost of energy thus affecting the operating cost of each material. Therefore, conducting an economic impact assessment involves referencing both the operational costs of similar plants and the most recent market prices for raw materials. In a real-world application, a financial officer would typically be on-site to account for these variables, and to optimize the profit of the site.

According to a review on performance standards of sulfuric acid plants done by the United States Environmental Protection Agency, a sulfuric acid plant in 1985 with a single absorber would cost roughly 1.23 million US dollars [31]. Adjusted for inflation, a plant of that size would cost roughly 3.6 million US dollars today. This price includes both direct (reactors, heat exchangers, piping, etc.) and indirect costs (engineering supervisor, contractor fee, etc.). As for operating costs, maintenance labor (employees) and operating work (ensuring plant functionality) cost roughly 11.89\$ / hr. and 10.89\$ / hr., respectively. When adjusted for inflation, maintenance labor is 35.34\$ / hr., and operating work is 32.37\$ / hr [32]. Small plants typically have up to 50 workers if each worker works around an 8-hour day, considering our plant is operational 8256 hours out of the year the total operational cost of the plant is \$3,489,432 USD \$4,742,440.32 CAD. It is important to note that this study was conducted in 1985, 5 decades ago. Sulfuric acid plants now cost much more, not just because of inflation but also because the materials used to build the plants are significantly more expensive. We have concluded that although this source is reliable, this price is simply too much of an underestimation. For example, a sulfuric acid plant in 2005, just 3 decades later would have cost 75 million USD [33].

Based on the most recent available data from mineral industry surveys, the cost of raw material sulfur for industrial use is approximately \$0.132 CAD/kg, as exported from Canada as reported January 2025. Assuming the plant operates with a sulfur consumption rate of 4.688 kg/s, and runs 8256 hours out of the year, the total sulfur consumption for the year would be approximately 39,019,955.2 kg/year resulting in a total cost of roughly 5.15 million CAD.

As of 2023 the market price for vanadium pentoxide is approximately \$3.33 CAD per kg. Assuming a typical catalyst load of 10,000 kg is used, results in an upfront cost of roughly \$33,300 CAD. Since the catalyst is typically replaced only once every 10 years, the annualized cost becomes approximately \$3,330 CAD per year [31]. For energy use, assuming the plant operates in Quebec and requires 128.66 MW of continuous cooling power over 8256 hours, the total yearly energy consumption is 1,062,079 MWh. With industrial electricity prices in Quebec ranging from \$200 to \$300 CAD per MWh, the estimated annual energy cost ranges between \$212.4 million and \$318.6 million CAD [32]. Thus, the total cost of the plant's operation is between 222.31 million CAD to 328.51 million CAD yearly. As for the revenue for the production of sulfuric acid, in April of 2024, sulfuric acid sold for \$0.14 USD per kilogram or \$0.19 CAD per kilogram [33]. Assuming that as per the assumption that 400,000 tons of sulfuric acid will be produced, this will result in a net revenue of \$76 million CAD. From the economic analysis, our plant is operating at a deficit of 151.05 million CAD to 257.25 million CAD. Much of the costs of the plant arise from the energy consumption, however, our plant unlike many other plants with exothermic reactions does not use superheated steam to generate internal energy. Recycling some of the lost heat may help to reduce the cost of energy and thus may be good improvement to implement. Optimization analysis as well may help the plant to increase revenue and better understand areas in which economic improvements may be made.

3.4 Environmental

The single-contact process for the production of sulfuric acid has two primary environmental concerns: off-gas emissions and the plant's energy consumption. Improvements that can be made in both areas that could potentially contribute to the plant's overall environmental sustainability. The plant undergoes a conversion rate of 97% of SO_3 to oleum. The 3% unreacted tail gas comprises 88% nitrogen, 0.002% unreacted sulfur dioxide, and 0.002% unreacted sulfur trioxide and 11.996% oxygen as calculated via our data. The unreacted sulfur dioxide and sulfur trioxide are the primary emissions that reduce the plant's environmental sustainability. The formation of SO_3 is due to incomplete absorption, and there are no known measures to reduce the percentage that is left unreacted. However, the conversion of SO_2 is dependent on working temperature, O_2/SO_2 ratio, and time rate of change of the SO_2 content in the reaction gas. The ideal inlet temperature of the catalyst bed to prevent equilibrium shift in the backward direction is 400–450°C [19]. For further improvements, process optimization analysis may be conducted. Each year, analysis may be conducted to determine the ideal working temperatures, pressure conditions, speed of reaction, and O_2/SO_2 ratio to maximize the conversion rate of SO_2 and limit greenhouse gas emissions.

As the plant expands production, it is feasible to consider a double-contact design as opposed to single-contact. The double contact process involves adding a second reactor and absorber to improve the conversion rate of SO_2 [34]. The production of the double contact process is more costly in the short term and requires more space; thus, it is not feasible for the current size of the plant; however, in the long term, it may be more economical as the conversion rate is between 99.82 and 99.85% as compared to the 97% conversion in the single contact process [19].

Another area that impacts the sustainability of the plant is energy consumption. Energy input into the system arrives from the high heat operation conditions for elements like the sulfur melter, the catalytic converter, and the furnace. Sulfuric acid production is an exothermic process; thus, a waste heat recovery system is implemented to improve energy efficiency. In modern sulfuric acid plants, up to approximately 67% of the process waste heat can be recovered and converted into superheated steam, which can then be used to generate on-site electricity to power systems. In the current design, major heat sources such as the sulfur furnace (R-101), catalytic converter (R-201), and absorption towers (R-301, R-302) offer opportunities for energy recovery through integrated heat exchangers. This allows the system to approach 85–90% thermal efficiency [19].

3.5 Improvements

Although our process is highly efficient and cost effective, there are improvements that can be made to boost our production. The main drawback of our process is the fact that it is a single contact process, rather than a double or higher contact process. Upgrading to a double contact process would involve the addition of a second absorption tower to maximize the conversion of sulfur trioxide into oleum (R-301). Roughly speaking, this will increase our conversion of sulfur dioxide from 97 to 99.7% [11]. While this would be a costly procedure, it would ultimately be worthwhile and would increase our conversion, and thereby our product by roughly 2.7%. Another way to increase productivity would be the addition of cyclone and gas cleaning units. These would take place after R-101 and R-201 and ensure that the product gas, either sulfur dioxide or sulfur trioxide, would be as pure as possible before entering the next phase. This would also help ensure

the longevity of both the vanadium oxide catalyst and the equipment, pipes, and processing units. Gas cleaning units would also help reduce our impact on the environment by ensuring we are not releasing any sort of physical debris into the air surrounding our plant [35]. Another improvement to our manufacturing system comes with the design itself. As shown in the process flow diagram, the initial air stream splits into two separate streams, fueling the reactions of both R-101 and R-201. Rather than split the streams, we could have instead provided a surplus (on top of the already double excess oxygen) of oxygen into stream 2 and simply have the unreacted oxygen in stream 4 fuel R-201. While this wouldn't necessarily bring down the total energy, it would reduce costs in building the plant as we would have one less stream to build. In the assumptions, it is assumed that in R-301, due to a temperature range of 50 to 70°C, no acid mist is formed. Realistically, a small quantity is produced which is why reactor such as the Hastelloy C-276 / C-2 is a necessary addition to reduce corrosion due to the acid mist [36].

3.5 Conclusion

The presented design project explores the industry production of sulfuric acid through the single contact process. Although the double contact process exists and is more efficient overall, it is assumed that our plant is relatively small and new to the production of said acid. With a 97% conversion rate of sulfur dioxide in the catalytic converter, compared to 99% and more for the double contact process, the overall process is nonetheless effective and efficient. The process starts off with extracting elemental sulfur from underground deposits using the Frasch process. This sulfur is liquified and stored. This is where the production begins. The contact process consists of 4 main reactors: a furnace to combust elemental sulfur into sulfur dioxide, a catalytic converter to oxidize sulfur dioxide into sulfur trioxide, an absorption tower to form oleum, and a dilution tower to dilute oleum into concentrated sulfuric acid.

At a high temperature of 431.15 K, 4.688 kg/s of sulfur is combusted with 40.222 kg/s of dry air to produce gaseous sulfur dioxide. The conversion is assumed to be 100%. The gas is fed into the catalytic converter alongside 19.47 kg/s of air to ensure surplus, only passing over the catalyst, vanadium oxide, once before being converted into sulfur trioxide gas. As previously mentioned, its conversion rate is 97%. The sulfur trioxide is cooled and introduced inside an absorption column in which a 13.775 kg/s stream of 98 wt% sulfuric acid reacts to form oleum. The oleum produced has a weight percent of 31% sulfur trioxide to 69% sulfuric acid which aligns with standard industrial percentages. The oleum is cooled down once more and is fed to a dilution column where a 2.755 kg/s supply of water dilutes the oleum into concentrated sulfuric acid where it is stored.

The overall process was assumed to produce 400,000 tons per year which includes a 3-week long period of downtime in which mechanics and technicians will ensure the facility, catalyst and apparatuses used in the plant is in good condition.

By working backwards using the conversions and yields found, a thorough mass and mole balance was created for each stream involved in the process. This leads into one of the most important considerations, heat balances. A heat balance was done around each of the 4 reactors discussed above. These energy outputs were calculated using the heat of formation method, utilizing hyperbolic heat capacity coefficient data for the species involved in each reaction. The sulfur furnace releases -30.696 MW of energy, the catalytic converter releases 11.973 MW of energy, the oleum absorption tower releases 50.0976 MW of energy and the oleum dilution tower releases 35.896 MW of energy, totalling at a summed value of 128.66 MW released. This large value requires lots of cooling to maintain the constant production of sulfuric acid. Although the estimated cost to run the plant would outweigh the calculated cost of electricity in Quebec. This can be resolved by implementing efficient equipment that does not consume so much electricity and possibly relocating.

4 List of Symbols

Symbol	Description
$\dot{n}$	Molar flow rate [mol / s]
$\dot{m}$	Mass flow rate [kg / s]
Q	Heat [MJ]
ΔH	Change in enthalpy [kJ]
ΔH_f^o	Heat of formation [kJ / mol]
dh	Infinitesimal change in energy [J / mol]
C_p	Heat capacity [J/ (mol K)]
T	Temperature [K]
P	Pressure [Bar]

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